

ENGINEERING CHEMISTRY LAB

I B. TECH I /II SEM: AERO, CSE, CSC, CSD, CSIT, CSM, EEE, ECE, IT, MECH								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS10	BSC	L	T	P	C	CIE	SEE	Total
		1	0	2	1.5	40	60	100
LIST OF EXPERIMENTS								
Experiment-1	Determination of total hardness of water by complexometric method using EDTA							
Experiment-2	Determination of Alkalinity of given water sample							
Experiment-3	Estimation of Chloride content of water by Argentometry.							
Experiment-4	Estimation of the concentration of an acid by Conductometry							
Experiment-5	Estimation of amount of ferrous ion by potentiometry using potassium dichromate							
Experiment-6	Estimation of HCl by potentiometry							
Experiment-7	Preparation of Bakelite							
Experiment-8	Preparation of Thiokol Rubber							
Experiment-9	Estimation of acid value of given lubricant oil							
Experiment-10	Estimation of Viscosity of lubricant oil using Ostwald's Viscometer.							
Experiment-11	Determination of rate of corrosion of mild steel in the presence and absence of inhibitor							
Experiment-12	Estimation of Surface tension of a given liquids using Stalagmometer							
VIRTUAL LABS								
Experiment-13	Construction of Fuel cell and its working							
Experiment-14	Smart materials for Biomedical applications							
Experiment-15	Batteries for electrical vehicles							
Experiment-16	Functioning of solar cell and its applications							
Reference Books:								
1. Lab manual for Engineering chemistry by B. Ramadevi and P. Aparna, S Chand Publications New Delhi (2022)								
2. Vogel's textbook of practical organic chemistry 5th edition								
3. Inorganic Quantitative analysis by A.I. Vogel, ELBS Publications.								
4. College Practical Chemistry by V.K. Ahluwalia, Narosa Publications Ltd. New Delhi (2007)								

Course Outcomes : Students should be able to:

The experiments will make the student gain skills on:

1. Determination of parameters like hardness of water and rate of corrosion of mild steel in various conditions.
2. Able to perform methods such as conductometry, potentiometry and pH metry in order to find out the concentrations or equivalence points of acids and bases.
3. Students are able to prepare polymers like bakelite and nylon - 6.
4. Estimations saponification value, surface tension and viscosity of lubricant oils.